Physics-Exact Credit Assignment for Multi-Agent Flutter Suppression

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Differentiating a structural plant's aeroelastic energy along its trajectories isolates a control-power term that is exactly additive across an arbitrary number of control inputs. We use this to give multi-agent reinforcement learning, applied to distributed flutter suppression on independently actuated surfaces of a Goland wing, a closed-form per-agent credit signal in place of the estimated one the method normally requires. Combined with log-energy shaping, this credit beats the closed form alone, a naive reward, and an adapted value-decomposition baseline, but not shaping alone once corrected for the full comparison family. Its advantage over the estimated alternatives is reliability, not raw mean performance, at higher variance. None of this was visible at a conventional exploration setting: the useful actuation amplitude is roughly forty times smaller than a conventional policy's exploration noise, and matching exploration to it changed performance by a factor of two. The credit signal tolerates plausible identification error, and a classical decentralised controller's fragility at the edge of the flight envelope is not a tuning artefact. The learned policies still trail that controller on average damping while holding one flight condition it cannot, pointing toward a certified, monitored-residual architecture not yet built.

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Nomenclature

$q, \dot{q}, \ddot{q}$	modal coordinates, rates, accelerations
M, C, K	modal mass, damping, stiffness matrices
L, x_{w}	wake-coupling matrix and wake (Wagner lag) states
B, δ, b_k	control-influence matrix, deflection vector, its kth column
E	instantaneous structural energy, $\frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$
P_k	control power of agent k, $(\dot{q}^\top b_k)\delta_k$
λ	energy decay rate, $\frac{1}{2}\,\mathrm{d}(\log E)/\mathrm{d}t$
D_k	difference reward for agent k
U, U_{f}	airspeed, flutter speed
N	number of control surfaces / agents
o_k, a_k	agent k's local observation and normalised action
v_r	rth in-vacuo mode shape, giving the modal residue $v_r^\top b_k$
σ	initial policy exploration standard deviation (normalised action units)
d, p	Cohen's d effect size, Welch's t-test p-value

I. Introduction

Flutter is a self-excited aeroelastic instability in which structural bending and torsion couple through unsteady aerodynamic loads and draw energy from the freestream; past a critical speed the response grows without bound within a few cycles, with no warning a pilot can act on. Because a wing must be certified against flutter with a substantial speed margin, the mass, stiffness and manufacturing cost of the structure are set in large part by an instability it is never meant to encounter in service [1, 2]. Active suppression buys back some of that margin with control effort rather than structural weight, and has attracted sustained engineering attention since the 1970s [3–5].

Two trends motivate revisiting the problem now. Wings are becoming more flexible and carry more independently actuated trailing-edge surfaces, each with its own local sensing on a bus rather than a dedicated wire to a central controller: the hardware is already distributed even where the control law is not. Reinforcement learning has also been applied to flutter suppression with encouraging results [6–8], but every instance we are aware of uses a single agent commanding a single effector from a global observation. Taking the hardware trend seriously means asking what happens when the effectors are commanded by separate agents that cannot see the whole wing, which is the question this paper answers on a specific, validated testbed (Figure 1).

Posing the problem with one agent per surface makes credit assignment the central difficulty, and for a physical reason: every surface acts on the *same* coupled bending–torsion mode, so a reward built from the whole wing's response says little about which surface helped. Standard multi-agent reinforcement learning remedies estimate that decomposition statistically from returns a policy happens to collect [9–11]. Our first observation is that this class of plant does not require estimation at all: for a structural system whose energy budget is exactly additive across control inputs, the credit decomposition is available in closed form from the equations of motion themselves.

Our second observation is the one we would most want a practitioner to take away. Every mechanism we tried – credit signals, communication schemes, action parameterisations, recurrence, auxiliary losses – plateaued in the same narrow performance band, evidence that something *upstream* of all of them was binding: the policy's exploration standard deviation was roughly forty times the deflection amplitude that actually damps this wing, so no credit signal, however well constructed, could assign credit to a signal buried that far under noise. Matching exploration to the useful actuation amplitude changed performance by a factor of two, more than any algorithmic choice we tried, and only then did the credit-assignment effect we set out to study become measurable at all. This echoes a known failure mode – the reinforcement learning literature has repeatedly found implementation and hyperparameter choices dominating algorithmic ones in general-purpose benchmarks [12–14] – but adds a diagnostic usable *before* training: the useful actuation amplitude is computable from a classical design, and the exploration scale can be checked against it in advance rather than discovered by trial and error.

Contributions

1) A closed form for the multi-agent difference reward in an energy-conserving structural plant with additive control authority (Proposition 1), with the correction that makes it usable: the credit must be the logarithmic decay rate, not the raw energy rate (Section IV).
2) A five-variant credit-assignment comparison, Holm-Bonferroni corrected, against a naive undecomposed reward and an adapted VDN-style value-decomposition baseline as well as each half of the closed-form-plus-shaping credit alone: the closed-form credit's advantage over an estimated one is consistency across seeds and conditions, not a uniformly larger effect size (Section V.B).
3) A quantified account of what limits learning on this problem, with a diagnostic usable in advance and a capability probe that rules out the competing explanation (Section V.D), plus a direct check of how much control-influence identification error the credit signal's sign tolerates (Section V.E).
4) A validated testbed and a six-rung classical baseline ladder that separates control method from information structure, cross-validated against an unsteady vortex-lattice solver, including a check of whether the classical baseline's fragility at the envelope edge is a tuning artefact (Sections III.A, V and V.A.1).
5) A negative result reported as such: the communication and action parameterisation mechanisms derived from

the physics do not help, at a level of statistical correction that makes the negative result harder to dismiss rather than easier.

II. Related work

The classical approach to active flutter suppression synthesises a single centralised regulator from a state-space aeroelastic model, driving all control surfaces from one observer and one gain [3–5]; Livne [1] and Mukhopadhyay [2] survey the field's history and the gap between laboratory demonstration and certified hardware. None of it questions the centralised structure itself: the regulator sees the whole state and commands every surface jointly, an assumption a physically distributed set of independently wired trailing-edge surfaces need not satisfy. A mature adaptive-control literature relaxes the model-fidelity assumption instead of the centralised one: Downs and Prazenica [15] and Mozaffari-Jovin et al. [16] adapt a single centralised regulator's parameters online, answering a different question from ours – robustness to model error at one control point, not what changes when the control points are separate decision-makers.

Deep reinforcement learning has also been applied directly to flutter suppression [6–8], but every instance shares the same structural assumption as the classical literature it replaces a hand-designed gain in: one learning agent, one effector, one global observation. A recent follow-up to the camber-morphing line distills a trained single-agent controller into a transparent policy via Shapley additive explanations [17] – a credit-assignment concept applied *post hoc* to interpret an already-fitted network, unlike our closed-form decomposition of the reward several agents train against *before* training begins. Shalymov et al. [18] frame a wing's surface as many small adaptively-controlled "feathers," which is multi-agent *control* in the classical sense rather than separate *learning* agents estimating from partial observability, the distinction that gives rise to the credit-assignment question this paper poses.

That decentralisation costs closed-loop performance relative to a centralised design with the same model is a classical result, not a symptom of using learning [19, 20], and Rotkowitz and Lall [21] show decentralised synthesis stays *tractable* only when the information structure is quadratically invariant – a condition our line-graph sensing pattern does not satisfy, which is why the fully decentralised LQG baseline (Supplemental Material, Section S2) is synthesised independently per surface rather than as one convex program. The standard tools for the credit-assignment problem we pose – difference rewards [22], learned value factorisation [9, 10], counterfactual baselines [11], surveyed in [23–25] – all estimate the credit decomposition statistically from returns a policy happens to collect, because the environment is treated as a black box; our contribution is narrower and complementary, exploiting physical structure that these general-purpose methods are built to work without. The formal setting is the decentralised POMDP [26, 27]. On learned communication, CommNet [28] and TarMAC [29] learn message content and routing end to end; we instead fix both to physics-derived, bandwidth-bounded choices, and report that neither helped (Section V.B).

Finally, our central caution about exploration scale has direct precedent in general-purpose reinforcement learning

[12–14, 30, 31], which has repeatedly found implementation and hyperparameter choices dominating algorithmic novelty. What we add is a case in which the dominant hyperparameter is *computable in advance* from a classical design of the same plant, which is not true of the hyperparameters that literature studies. No prior study we are aware of combines multi-agent treatment of a wing's individual control surfaces, a closed-form (rather than estimated) credit signal, and validation of the plant against a higher-fidelity solver; that combination, and the negative and cautionary results it surfaces, is this paper's contribution (Supplemental Material, Section S4).

III. Problem formulation and plant model

We model the task as a Dec-POMDP [26, 27] with one agent per control surface. The state $s = (q, \dot{q}, x_{\mathrm{w}}, \delta)$ collects the modal coordinates, rates, wake states, and current deflections; agent k chooses a normalised command $a_k \in [-1, 1]$, converted by its actuator to a physical deflection subject to a rate limit and travel stops. Each agent observes only o_k: an accelerometer pair at its own station, local plunge and twist rate, its own saturation fraction, and air data – crucially, not the modal state. The unstable mode is a global quantity no single station can measure, so partial observability is intrinsic to the physical arrangement rather than an artefact of the formulation. Communication, where used, is nearest-neighbour along the span. We use the standard discounted formulation [32]; the quantity to be minimised, and the quantity we report, is the exponential growth rate of the structural energy E,

$$\lambda \;=\; \tfrac{1}{2}\,\frac{\mathrm{d}}{\mathrm{d}t}\log E, \tag{1}$$

negative when the wing is being damped. Stating the objective this way matters later: it lets the per-agent credit and the evaluation metric be literally the same quantity (Section IV).

A. Model

We use a modally reduced uniform cantilever wing (Figure 1). Three trailing-edge surfaces occupy spanwise bands at $[0.05, 0.40]$, $[0.40, 0.72]$ and $[0.72, 0.98]$ of the semispan, hinged at 75 % chord, with $\pm 15^\circ$ travel, $200\,^\circ\,\mathrm{s}^{-1}$ rate limit, and a 20 ms first-order servo lag; control runs at 200 Hz. The structure is a Galerkin projection onto exact clamped-free bending and fixed-free torsion modes; aerodynamics are strip theory carrying the full Theodorsen non-circulatory terms [33], with circulatory lag realised by two Wagner lag states per strip [34], giving

$$M\ddot{q} + C\dot{q} + Kq \;=\; L\,x_{\mathrm{w}} + B\delta + b_{\mathrm{g}} w_{\mathrm{g}}, \tag{2}$$

a 56-dimensional state: four modal coordinates and rates, plus two wake states on each of 24 strips. Full parameters are tabulated in the Supplemental Material (Section S3) for reproducibility. The model agrees with the published Goland cantilever wing [35] to better than 0.5 % on modal frequencies and flutter speed (Table 1), an automated regression test

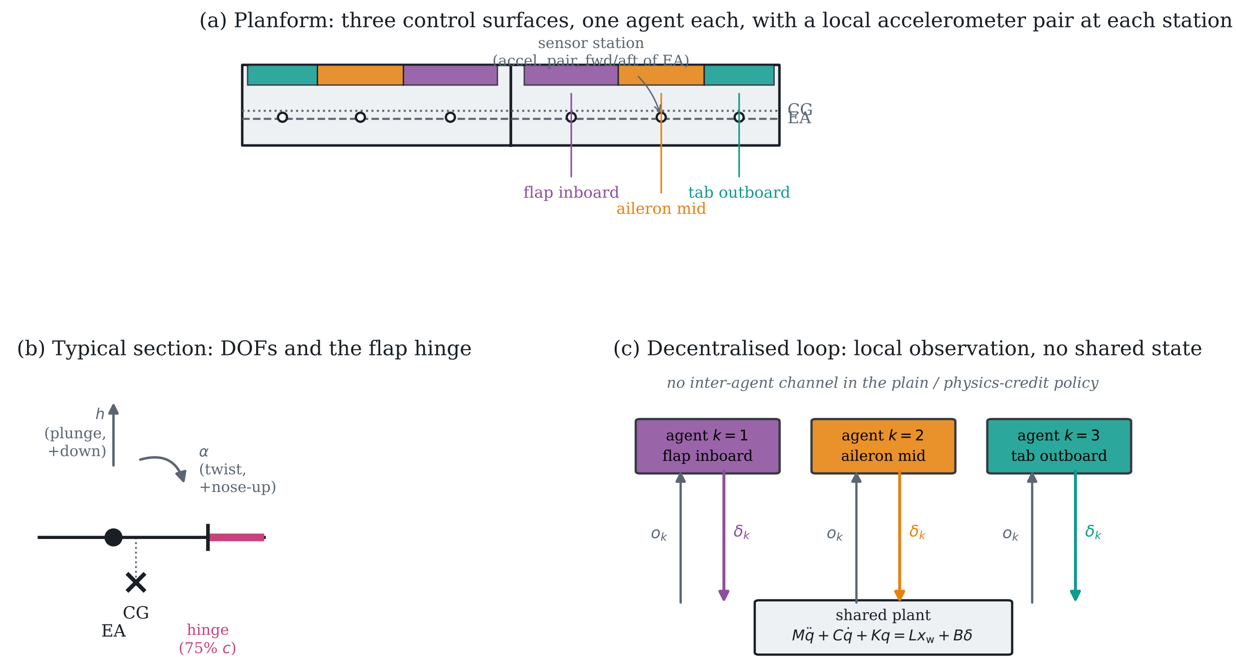

Fig. 1 The testbed. (a) Planform, agent colours matched throughout the paper. (b) The typical-section degrees of freedom and flap hinge. (c) The information structure: each agent closes its own loop through the shared plant, with no inter-agent channel in the plain and physics-credit policies.

rather than a one-time check.

Table 1 Validation against the published Goland wing.

Quantity	This model	Published
First bending frequency	7.664 Hz	7.66 Hz
First torsion frequency	15.232 Hz	15.24 Hz
Flutter speed	137.35 m s^{-1}	$\approx$ 137 m s^{-1}
Flutter frequency	69.3 rad s^{-1}	$\approx$ 70.7 rad s^{-1}
Static divergence speed	356.8 m s^{-1}	—

B. Cross-validation against an unsteady vortex-lattice solver

The model buys speed with two approximations a reader is entitled to distrust: two-dimensional strip aerodynamics, and quasi-steady flap derivatives that neglect the flap's own shed wake. We quantified both against SHARPy [36], coupling an unsteady vortex-lattice method [37] to a geometrically exact beam with structural data identical to ours, so any disagreement is about the aerodynamics. Flutter frequency agrees to 0.06 %; the critical *speed* differs by −11.1 %, the direction strip theory predicts (no tip relief, so it over-predicts loading and under-predicts flutter speed) – our plant is conservative, the safe direction for a control study. Control authority runs the other way and is larger: a part-span flap loses effectiveness to spanwise carryover a sectional derivative cannot represent, and our plant over-estimates whole-wing control authority by a factor of 1.9 relative to UVLM (grid-converged to within 3 %; full derivation in the Supplemental Material, Section S3). Every controller in this study receives the same inflated authority, so relative comparisons and significance tests are unaffected; absolute decay rates would not transfer unchanged to a UVLM plant, and we do not claim they would. The two discrepancies act in opposite directions – a harder flutter problem, an easier control problem – so they cannot be combined into a single safety factor, but the control-authority number recurs twice more below as an independent check on other findings.

IV. Method

Differentiating the aeroelastic energy $E = \frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$ along trajectories of (2) gives

$$\dot{E} = \underbrace{-\dot{q}^\top C\dot{q}}_{\text{dissipation}} + \underbrace{\dot{q}^\top Lx_{\mathrm{w}}}_{\text{wake exchange}} + \sum_k \underbrace{(\dot{q}^\top b_k)\,\delta_k}_{P_k}, \tag{3}$$

with b_k the kth column of B. The final term is exactly additive across agents; $P_k < 0$ means surface k removes energy from the structure.

Proposition 1 (Closed-form difference reward)**.** *For a global objective whose only agent-dependent term is the control power in* (3)*, the difference reward $D_k = G(z) - G(z_{-k})$ admits the closed form $D_k = -P_k$. No counterfactual rollout, learned mixing network, or centralised critic is required to evaluate it.*

Proof. Removing agent k sets $\delta_k = 0$. Every other term of (3) is independent of δ_k, and the control-power term is a sum over agents, so the difference collapses to $(\dot{q}^\top b_k)\delta_k$. □

Projecting onto the in-vacuo modes keeps the separation in agent *and* mode: agent k's contribution to mode r is $\dot{\eta}_r(v_r^\top b_k)\delta_k$, where the modal residue $v_r^\top b_k$ decides whether a surface damps or excites that mode and whose sign inverts under control reversal – the quantity Section V.E stress-tests against identification error.

Used directly, $-P_k$ fails instructively: it is maximised by having energy available to extract, so an agent is rewarded for *sustaining* the oscillation. Policies trained on it held the wing at a bounded but undamped amplitude, scoring $+0.265\ \mathrm{s}^{-1}$ against -0.06 for a hand-tuned collocated damper – reward hacking, not myopia. Dividing (3) by E repairs it,

$$\frac{\mathrm{d}}{\mathrm{d}t}\log E = \frac{-\dot{q}^\top C\dot{q} + \dot{q}^\top Lx_{\mathrm{w}} + \sum_k P_k}{E}, \qquad r_k = -\frac{P_k}{E+\varepsilon}, \tag{4}$$

and because E is a shared scalar the decomposition survives division exactly, so Proposition 1 carries over to the log-decay rate: scale invariant, and coinciding with the evaluated objective (1). Making this change inverted the sign of the outcome, from $+0.265$ to $-0.291\ \mathrm{s}^{-1}$, with training divergence falling from 0.8 % to 0.0 %.

Remark 1. *The exact signal remains myopic: cross-mode coupling through C and the wake means an agent may have* *to inject energy into one mode so another can extract more from the coupled pair. We add potential-based shaping [38] on* $\log E$*, with a shaping discount of one rather than the RL discount – a discount below one leaves a constant negative leakage reward once the wing is damped (*-6.91 *per step,* -977 *over an episode for a perfect controller), which an invariance theorem cannot excuse if it ranks a ringing wing above a damped one.*

We use shared-parameter multi-agent PPO [31, 39] with generalised advantage estimation [40] and per-agent advantages, on-policy because the plant is cheap to simulate and the per-agent credit is recomputed exactly at every step rather than replayed. Updates run on sequences of consecutive steps with a detached stored hidden state; sampling shuffled individual timesteps, as we did initially, gives a recurrent encoder no temporal gradient at all [41] – one illustrative run under the old sampler reached $-0.213\,\mathrm{s}^{-1}$ against -0.566 ± 0.123 for the same variant under the sequence-based sampler used everywhere else. Full training pseudocode, hyperparameters, and network architecture (the trunk, an optional recurrent phasor encoder, optional spanwise-line consensus, and an optional phase-locked action head – the mechanisms ablated in Section V.B under the names `recurrent_only`, `mocca_nohead`/`mocca`, and `comm_raw`) are given in the Supplemental Material (Section S1) to stay within the journal's length guidance. Every head's final layer is scaled by 0.01 at initialisation so a fresh policy is close to a no-op, which is what makes the residual variants of Section V.A well-posed from the first update. Actions are Gaussian with a learned, state-independent log-standard-deviation shared across agents – the quantity swept in Section V.D.

V. Experimental evaluation

A distributed controller cannot be judged against a centralised one without confounding the control method with the information available to it, so we report six rungs: collocated local rate feedback (no model, no communication); gain-scheduled LQR on the full 56-state plant (not implementable, a ceiling); centralised LQG on the same six accelerometer channels the agents see; a fully decentralised LQG in which each surface runs its own observer on its own two accelerometers and commands only itself [19, 42] – the honest target for a distributed policy; and the same design re-tuned for control-authority margin (Section V.A.1). All are synthesised in discrete time at the control rate (synthesis equations in Supplemental Material, Section S2). A fault-aware oracle – a gain bank re-synthesised per failure hypothesis, given the full state and told which surface failed – checks feasibility rather than competing: of nine evaluation conditions, 6 are stabilisable by any controller with these actuators, and the rest are reported as infeasible rather than as failures of a method. Every controller is scored on the same physical quantities over identical episodes – decay rate (1), peak tip plunge, RMS deflection, divergence fraction – since PPO return cannot rank runs whose reward functions differ; following [30] we report effect sizes and significance rather than means alone. Training-time convergence behaviour for every variant is given in the Supplemental Material (Section S1); every variant reaches a stable plateau well inside the training budget.

Table 2 Energy decay rate by controller and flight condition. Negative is suppressed, `D` marks the fraction of episodes that diverged, and – marks conditions infeasible for any controller.

Condition	local_fb	lqr_fixed	lqr_sched	lqg_local	dlqg_local	dlqg_robust	lqr_oracle	comm_raw	ippo_blended	ippo_physics	mocca_nohead	mocca	recurrent_only	residual_only	shaped_shared
U=1.05Uf healthy	-0.06	-4.40	-4.61	-4.71	-5.42	-4.17	-4.54	-0.78	-0.99	-0.90	-0.68	-0.85	-0.82	-1.98	-0.96
U=1.05Uf jam#2@8deg	-0.10	-0.10	-0.12	-0.13	-0.09	-0.11	-0.06	-0.05	-0.06	-0.06	-0.08	0.01	-0.08	-0.07	-0.10
U=1.05Uf jam#2@14deg	-0.12	-0.09	-0.10	-0.11	-0.08	-0.10	-0.06	-0.06	-0.08	-0.07	-0.06	D42%	-0.08	-0.07	D33%
U=1.25Uf healthy	-0.01	-5.27	-4.93	-5.38	-1.07	-0.82	-4.93	-0.62	-0.96	-0.77	-0.46	-0.47	-0.51	-1.92	-0.65
U=1.25Uf jam#2@8deg	D100%	D100%	D50%	D100%	D50%	D50%	-0.12	D75%	D100%	D100%	D100%	D100%	D100%	D50%	D100%
U=1.25Uf jam#2@14deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U=1.45Uf healthy	D100%	-5.79	-5.68	-6.45	0.32	0.18	-5.74	-0.39	-1.30	-0.03	0.21	0.15	-0.09	-0.69	-0.47
U=1.45Uf jam#2@8deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U=1.45Uf jam#2@14deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Stabilised	4/6	5/6	5/6	5/6	5/6	5/6	6/6	5/6	5/6	5/6	5/6	4/6	5/6	5/6	4/6

A. Results

Three things in Table 2 matter. First, the classical ladder separates by information structure rather than sophistication: centralised LQG reaches $-5.52\,\mathrm{s}^{-1}$ on healthy conditions, the fully decentralised LQG (seeing exactly what the agents see) reaches -2.05 – a factor of 2.7 cost from decentralisation, consistent with the classical expectation [19, 21] – which is why the decentralised controller, not the centralised one, is the reference for a distributed policy. Second, the best learned policy reaches $-1.53\,\mathrm{s}^{-1}$, about three quarters of the decentralised classical result; the average conceals more: broken out by condition the decentralised LQG achieves -5.42, -1.07 and $+0.32$ at 1.05, 1.25 and 1.45 U_f – failing outright at the top of the envelope – against -1.98, -1.92 and -0.69 for the learned policy, which is far more uniform across the envelope and holds a condition where the classical one diverges, while losing badly where the classical one is strongest. Third, and least comfortable: a robustness advantage measured at a conventional exploration scale does not survive at the scale that maximises damping. At $\sigma = 0.30$ the learned policies held all six feasible conditions, matching the fault-aware oracle; at $\sigma = 0.05$ they hold five, as the classical controllers do, while damping three times better. Exploration scale trades robustness against nominal performance; we report both halves.

1. Is the classical baseline's fragility a tuning artefact?

An unstructured LQG has no guaranteed stability margin, unlike full-state LQR [43], so before reading anything into the learned policy holding a condition the decentralised LQG diverges on, we checked whether a better-tuned classical design closes the gap. Loop transfer recovery [44] – the standard remedy for an observer-induced margin loss – does not help: boosting the Kalman filter's process-noise weighting across four orders of magnitude leaves the control-authority margin at 1.45 U_f unchanged (0.158 throughout), because the fully state-fed loop already sits at the same margin. A 50× increase in the LQR effort weight does move it, from 0.158 to 0.179 before plateauing, at a measurable cost to nominal decay rate; the re-tuned design (`dlqg_robust` in Table 2) improves the 1.45 U_f growth rate from $+0.32$ to $+0.18\,\mathrm{s}^{-1}$ – smaller, but still growing – while giving up performance elsewhere (-0.82 against -1.07 at 1.25 U_f). Neither classical

Table 3 Credit-assignment ablation at $\sigma = 0.05$, six seeds per variant. All ten pairwise p-values are Holm-Bonferroni corrected across this family.

Variant	Seeds	Decay rate [1/s]	Conditions held
`ippo_blended`	6	-1.085 ± 0.243	5.0/9
`ippo_physics`	6	-0.566 ± 0.123	5.0/9
`ippo_shared`	6	-0.526 ± 0.806	4.0/9
`shaped_shared`	6	-0.690 ± 0.145	4.3/9
`vdn_shared`	6	-1.670 ± 0.699	1.8/9

Comparison		Cohen's d	Hedges' g	p	p_{Holm}
`ippo_blended`	vs `ippo_physics`	−2.69	−2.49	0.0020	0.0198*
`ippo_blended`	vs `ippo_shared`	−0.94	−0.87	0.1557	0.5683
`ippo_blended`	vs `shaped_shared`	−1.97	−1.82	0.0088	0.0795
`ippo_blended`	vs `vdn_shared`	1.12	1.03	0.0996	0.4980
`ippo_physics`	vs `ippo_shared`	−0.07	−0.06	0.9078	1.0000
`ippo_physics`	vs `shaped_shared`	0.92	0.85	0.1421	0.5683
`ippo_physics`	vs `vdn_shared`	2.20	2.03	0.0112	0.0895
`ippo_shared`	vs `shaped_shared`	0.28	0.26	0.6426	1.0000
`ippo_shared`	vs `vdn_shared`	1.52	1.40	0.0257	0.1543
`shaped_shared`	vs `vdn_shared`	1.94	1.79	0.0177	0.1240

design holds the condition the learned policy holds. This margin (0.158–0.179) is also comfortably clear of the plant's own 1.9× control-authority overestimate (Section III.B), a safety factor of roughly 3× in the direction that already occurred: the fragility at the envelope edge is structural to decentralised control here, not explained away by tuning or by the plant's own modelling error. Full sweep data are in the Supplemental Material, Section S2.

B. Ablations

Credit assignment Six seeds per variant at $\sigma = 0.05$ (Table 3), five variants: the exact-plus-shaping combination (`ippo_blended`), each half alone (`ippo_physics`, `shaped_shared`), the naive undecomposed shared reward (`ippo_shared`), and a learned value-decomposition credit adapted from VDN [9] (`vdn_shared`): each agent keeps its own state-value head, but the *summed* value is fit to the team return and every agent's update uses that shared advantage, decomposing the value function as VDN's architecture does without reproducing its discrete-action mechanism. Five variants means ten pairwise comparisons, so every p-value here and in Table 3 is Holm-Bonferroni-corrected across that full family, with effect sizes as Hedges' g (small-sample-corrected) alongside Cohen's d.

The combination reaches -1.085 ± 0.243, beating the closed-form credit alone (-0.566 ± 0.123) at $g = -2.49$, $p_{\text{Holm}} = 0.020$ – the one comparison here that survives correction. It does *not* significantly beat shaping alone (-0.690 ± 0.145, $g = -1.82$, $p_{\text{Holm}} = 0.080$): tested alongside only two other variants, that comparison's uncorrected $p = 0.009$ looks significant, but does not survive the five-variant family the physical question calls for. The naive shared reward averages -0.526 ± 0.806 – a spread three to six times wider than any credit-bearing variant's, indistinguishable

Table 4 Architecture ablation at $\sigma = 0.05$. Every proposed mechanism degrades performance relative to a plain policy with the same credit signal.

Variant	Seeds	Decay rate [1/s]	Conditions held
`comm_raw`	6	-0.596 ± 0.227	5.0/9
`mocca`	6	-0.389 ± 0.219	4.2/9
`mocca_nohead`	6	-0.308 ± 0.169	5.0/9
`recurrent_only`	6	-0.474 ± 0.249	5.0/9

Comparison		Cohen's d	Hedges' g	p	p_{Holm}
`comm_raw`	vs `mocca`	−0.93	−0.86	0.1391	0.6953
`comm_raw`	vs `mocca_nohead`	−1.44	−1.33	0.0336	0.2018
`comm_raw`	vs `recurrent_only`	−0.51	−0.47	0.3961	1.0000
`mocca`	vs `mocca_nohead`	−0.41	−0.38	0.4907	1.0000
`mocca`	vs `recurrent_only`	0.36	0.33	0.5444	1.0000
`mocca_nohead`	vs `recurrent_only`	0.78	0.72	0.2104	0.8414

from all three after correction ($p_{\text{Holm}} \geq 0.57$): not an outright failure, but far less reliable. The value-decomposition credit is the opposite failure mode: its mean, -1.670 ± 0.699, is the most negative of all five ($g = -2.03$ against the closed form alone, $p_{\text{Holm}} = 0.090$, short of significance but the largest point estimate), while its variance is nearly three times `ippo_blended`'s and it stabilises only 1.8 of nine conditions on average against 5.0 for every credit-bearing variant – damping harder where it stabilises at all, at the cost of stabilising far fewer conditions. These are the two failure modes a learned credit signal can have when the environment does not hand it one: no decomposition gives high-variance performance; a decomposition estimated by summing per-agent value heads to a team target gives aggressive but inconsistent performance. The physics-derived signal is the only one whose seed-to-seed spread and stabilised-condition count are both consistent across seeds, a property worth weighing alongside the mean.

Architecture None of four physics-motivated architectural mechanisms survives measurement (Table 4). A plain feedforward policy with the blended credit reaches -1.085 ± 0.243; adding a recurrent phasor encoder gives -0.474 ± 0.249, adding spanwise-line consensus -0.308 ± 0.169, adding a phase-locked action head -0.389 ± 0.219, and sharing raw neighbour observations -0.596 ± 0.227 – all four significantly worse after Holm-Bonferroni correction ($g = -2.29$ to $g = -3.43$ for the first three, $g = +1.92$ for raw sharing, all $p_{\text{Holm}} \leq 0.034$). We do not read this as settling whether communication helps distributed flutter suppression in general: three surfaces on a six-metre wing, with air data already broadcast, is a small coordination problem, and at eight surfaces the picture changes qualitatively (Supplemental Material, Section S5). It does settle that on this problem the credit signal was worth deriving from the physics and the message content was not.

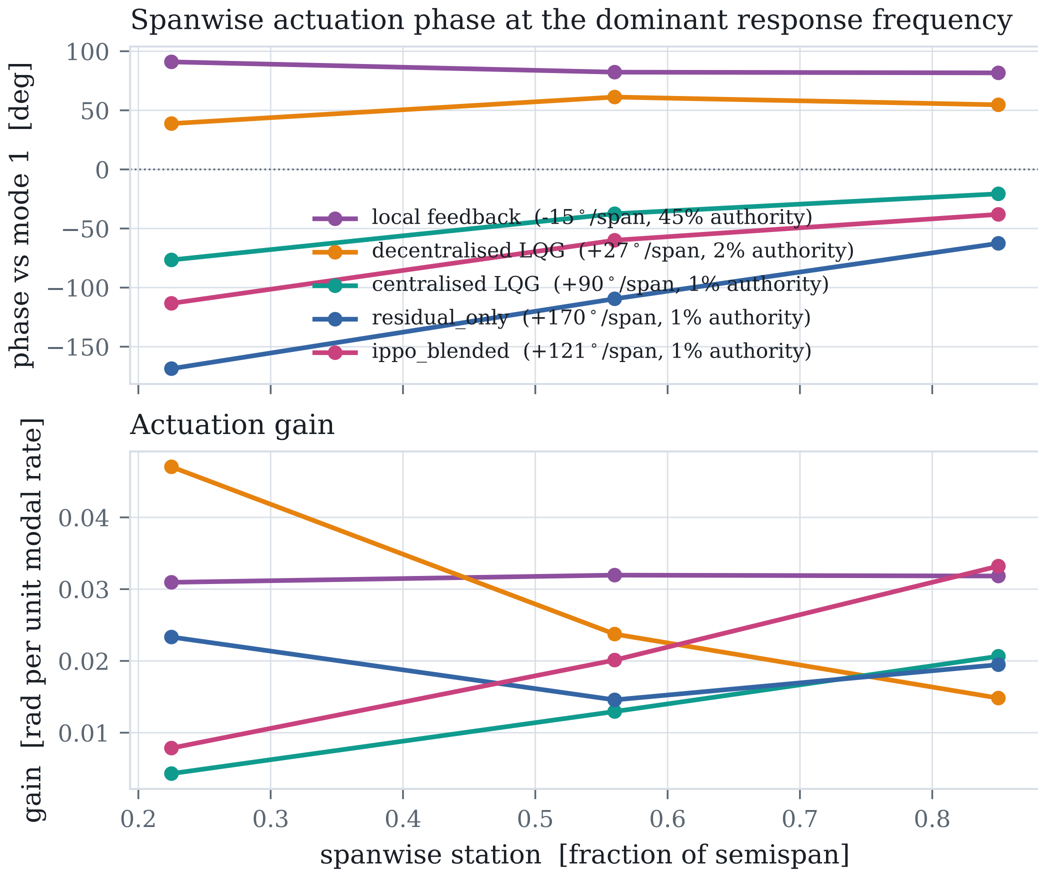

Fig. 2 Spanwise actuation phase (top) and gain (bottom) at each controller's dominant response frequency. A sloped phase profile is a travelling-wave pattern matched to the mode shape; a flat profile is purely local reaction.

C. Emergent spanwise phase coordination

"Emergent coordination" is easy to assert and hard to pin down, so we define it as something measurable: the phase at which each surface deflects relative to the critical mode's velocity, by spanwise station, measured identically for every controller via the cross-spectrum of each surface's deflection against the first bending mode's velocity at the frequency where the modal response peaks.

Applied to the classical ladder the instrument separates the rungs sharply: local rate feedback is essentially flat ($-15°$ per semispan at 5.50 Hz) and never engages the unstable mode at all; the fully decentralised LQG reaches $+27°$ at 7.50 Hz; centralised LQG develops $+90°$ at 11.00 Hz, matching the 11.03 Hz eigenvalue prediction almost exactly – the more capable the controller, the stronger its spanwise phase progression. The learned policies give this a genuine test: the strongest credit-only policy, `ippo_blended`, locks onto 11.00 Hz with a phase gradient of $+121°$, steeper than the centralised design's, from six accelerometer channels alone with no communication and no information about the mode shape beyond what the reward and dynamics implicitly encode. The residual policy instead locks onto a different, higher-frequency mode ($+170°$ per semispan at 15.50 Hz) rather than the primary flutter mode, consistent

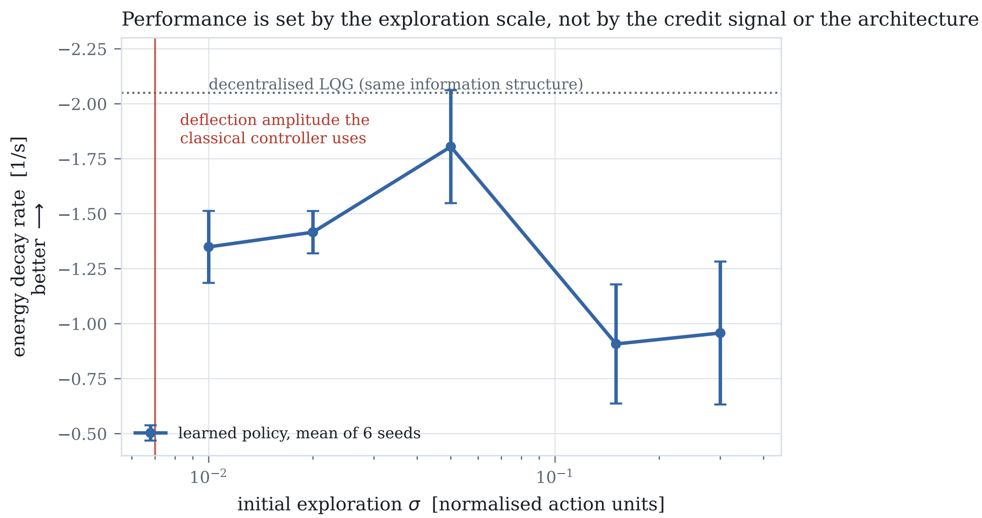

Fig. 3 Decay rate against initial exploration standard deviation, all else fixed. The vertical line marks the deflection amplitude the classical controller uses. Six seeds per point.

with it correcting a base law that already suppresses the primary mode and spending its own authority on what is left. We read the credit-only result as the concrete, falsifiable content behind "emergent" in this paper: a distributed policy that has never been told the mode shape converges, from local measurement alone, on the same resonant frequency full-state, full-model synthesis converges on by construction. The same measurement corrects a wrong inference authority alone would suggest: local rate feedback uses 44.7 % of available deflection to reach $-0.03\,\mathrm{s}^{-1}$, centralised LQG uses only 0.7 % to reach −5.52 – the better the controller, the *less* authority it spends, because flutter suppression is a phase-precision problem rather than a control-power one, and the learned policies (1.2 % to 1.3 % authority) apply authority comparable to the good classical controllers, at a phase closer to optimal than local feedback but not as sharp as the centralised design.

D. What limits learning here

This is the finding with the widest reach. Holding everything else fixed and varying only the initial exploration standard deviation moves performance by a factor of two (-1.805 ± 0.257 at $\sigma = 0.05$ against -0.908 ± 0.271 at $\sigma = 0.15$, a ratio of 1.99), with an interior optimum at $\sigma = 0.05$. The mechanism reads off the axis: a controller achieving the classical result uses about 0.105° RMS deflection; $\sigma = 0.30$ corresponds to 4.5° – the policy's own exploration is some forty times the amplitude that works, and no credit decomposition can recover a signal buried underneath it. This is not a tuning remark: before finding it, every variant plateaued between −0.03 and $-0.82\,\mathrm{s}^{-1}$ regardless of credit signal, communication, action parameterisation, memory, auxiliary supervision, or base controller (six-seed variance details in the Supplemental Material, Section S5). A capability probe rules out the competing explanation: training on the single easiest condition gives +0.21 and +0.27, worse than training on the full distribution, so the plateau was never a matter of the task being too broad. The practical recommendation is specific: the useful actuation amplitude is computable before any training, from a classical design or an estimate of the damping authority required. Scale the exploration to it; on this problem that single number mattered more than every algorithmic choice we made, and it is cheap to check.

E. Sensitivity to control-influence identification error

The credit signal's sign is decided by the modal residue $v_r^\top b_k$, free in simulation but requiring an identified control-influence estimate on hardware; a wrong sign rewards a surface for exciting a mode rather than damping it. Modelling identification error as an independent relative perturbation on each entry of B, the most fragile surface–mode pairing flips sign at just 11 % relative error, essentially speed-invariant across the flight envelope; every other entry tolerates more, up to entries that do not flip within ±100 % error at all (full per-entry results in the Supplemental Material, Section S6). Under a *uniform* per-surface scale error instead, every sign flips simultaneously only at −100 % (classical control reversal); the plant's own +90 % UVLM control-authority overestimate (Section III.B) moves the assumed value *away* from that boundary, a safety factor of roughly 2–3× in the direction that already occurred. Retraining with the credit signal's belief about control authority perturbed by a uniform ±60 % (three seeds) does not collapse performance or reverse its sign in either direction (Supplemental Material, Section S6): moderate uniform identification error degrades gracefully rather than catastrophically, though we have not tested past the −100 % reversal boundary itself.

VI. Discussion and conclusion

This study set out to ask whether treating each control surface as a separate learning agent is workable for flutter suppression, and the answer is conditional. On credit assignment, yes, with numbers behind it: a physics-derived signal, requiring no estimation step, is competitive with the estimated alternatives we tested and more consistent across seeds and conditions than any of them. On whether the resulting policies are ready to fly, not yet – the gap to a classical decentralised design on average nominal damping is real rather than an artefact of tuning (Section V.A.1), and the learned policies have no certification pathway or stability-margin guarantee, unlike every classical rung.

The most actionable finding for practitioners is the least glamorous one: before training a policy for a physical control task, compute the actuation amplitude a competent classical controller would use and check the initial exploration scale against it. On this plant the two were separated by a factor of forty at a conventional PPO setting, and every algorithmic refinement we tried was invisible until that gap was closed; we expect this check to be worth running on any task where a classical baseline is available even loosely. Flutter suppression is flight-safety-critical by construction, and nothing here argues the learned policies as trained are ready for that role – we place this work at approximately TRL 2–3, a validated simulation testbed and a first-principles credit signal with no wind-tunnel or hardware-in-the-loop evidence yet. The residual formulation offers the most concrete path we see toward certifiable: `residual_only` is, by construction, a certifiable decentralised LQG base law plus a learned correction initialised near zero and bounded to a fixed authority fraction. A run-time-assurance architecture on the same structure – the certified law as default, the correction bounded, and a monitor that reverts to the base law outside a pre-verified envelope – would let the learned component fail safe into a controller already certifiable on its own; we have not built or verified such a monitor, nor characterised how the credit signal's identification-error robustness interacts with the monitor's authority bound, and flag both as the concrete next step.

We did not expect the phasor consensus and phase-locked action head to hurt. A plausible interpretation is that a three-surface, six-metre wing with broadcast air data is too small a coordination problem for an explicit channel to earn its bandwidth cost, and the eight-surface result (Supplemental Material, Section S5) is at least consistent with that story, though not run to the statistical standard we hold the main claims to; we read this as a caution against building communication machinery into a distributed controller by default, without first checking whether a plain policy with the same credit signal already does the job. We still find the phase-coordination result the most striking one: a policy trained end to end on a shaped scalar reward, using only its own two accelerometers, converges on the same resonant frequency that a full-state, Kalman-filtered LQG converges on by construction – a property of the coupled structure the policy has to find, not supplied anywhere in the reward, the observation, or the network architecture, which is the sense in which we think "emergent" is earned here rather than merely asserted.

A. Limitations

- The learned policies do not match the decentralised classical controller on average nominal damping (-1.53 against $-2.05\,\mathrm{s}^{-1}$); they are more uniform across the envelope and hold one condition it diverges on, but we do not claim they dominate it.
- The robustness result is scale-dependent (Section V.A); we report the trade-off rather than its better half.
- The plant over-estimates control authority by $1.9\times$ relative to UVLM (Section III.B); relative comparisons are unaffected but absolute decay rates would not transfer, and we have not retrained on the UVLM plant or characterised how the credit decomposition would degrade under a higher-fidelity model with inter-surface aerodynamic coupling.
- The credit signal requires a modal control-influence estimate that is free in simulation but identified on hardware; the least robust surface–mode pairing tolerates only 11 % independent relative error before its sign reverses (Section V.E), though the plant's own measured discrepancy is well clear of that boundary.
- Ablations use six seeds on a three-surface wing; the eight-surface results and three further probes (Supplemental Material, Section S5) are one- and two-seed and not run to the same standard. Correcting for the number of comparisons made changed which credit comparisons are significant, and bringing the exploration sweep to six

seeds changed our estimate of its seed-to-seed spread substantially, though not the qualitative shape of the curve.

- The exploration finding is established on this plant and algorithm; we expect it to generalise to control tasks with a narrow useful actuation band relative to the action range, but have not shown that.

B. Conclusion

Posing flutter suppression as a distributed learning problem exposes a credit structure the physics resolves in closed form: the difference reward is an agent's own control power, normalised by the instantaneous energy so it coincides with the objective and cannot be gamed by sustaining the oscillation. Combined with log-energy shaping it beats the closed form used alone, once compared and corrected for multiple comparisons against the full family of alternatives a reviewer would want to see it measured against; against estimated credit signals, its advantage is reliability across seeds and conditions rather than a uniformly larger effect. The more transferable lesson is the caution: that result, and a clean negative result on communication machinery, were both invisible until the exploration scale was matched to the actuation amplitude that damps the wing – a quantity computable in advance from a classical design, which mattered more here than any algorithmic choice we made, and which we suspect is true of active vibration control more broadly.

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Data and Code Availability

The plant, baselines, training code, evaluation harness, and the SHARPy cross-validation and sensitivity-analysis scripts referenced throughout this paper and its Supplemental Material are available at the project repository.

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